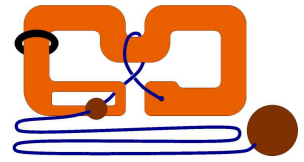




SHORTLIST

2019

INTERNATIONAL MONSTERS' OLYMPIAD



The Real Shortlisted Problems



Problem Selection Committee:



Johnny Bedevere Arthur (chair) Black Knight Bridgekeeper

2nd International Monsters' Olympiad
Bath — UK, 11th–22nd July 2019

Problems

Algebra

A1. Let $\mathbb{Z}$ be the set of integers. Determine all functions $f: \mathbb{Z} \rightarrow \mathbb{Z}$ such that, for all integers a and b ,

$$f(2a) + 2f(b) = f(f(a + b)).$$

A2. When the age of Ann will be the same as Mary's age now, Mary will be exactly 32 years older than Ann was when Mary was half as old as Ann is now. How old can be Mary?

A3. **Arthur Dumpling** (Hungarian cartoon character: a fat bird who loves chocolate of all kinds) lives at 1 Sweet Street. The chocolate shop is operating at number n , the far end of the street. Arthur's daily fitness programme is as follows: he starts in front of number 2. When he stands in front of number k (where $1 < k < n$), he tosses a fair chocolate coin. If it shows heads, he moves to number $(k - 1)$. If it shows tails, he moves to number $(k + 1)$. If he reaches the chocolate shop, he enters and throws a chocolate ball down his throat, and then moves to number $(n - 1)$. If he arrives back home, the fitness programme terminates. On average, how many chocolate balls does Arthur throw down his throat per day?

A4. Show that for every odd integer $N > 5$ there exist vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in (three-dimensional) space which are pairwise perpendicular, not parallel with any of the coordinate axes, have integer coordinates, and satisfy $|\mathbf{u}| = |\mathbf{v}| = |\mathbf{w}| = N$.

A5. Does there exist an infinite sequence $a_1, a_2, \dots$ of real numbers which is bounded, not periodic, and satisfies the recursion $a_{n+1} = a_{n-1}a_n + 1$?

A6. Let $P(x)$ be a polynomial of degree 100 with real coefficients such that $P(x) - 1$ has 65 distinct real roots, and $P(x) + 1$ has 85 distinct real roots. Determine the minimal possible number of roots of $P(x)$.

A7. King Arthur thinks of three positive real numbers. He writes down on an oak leaf, in some order, six numbers which are pairwise products and pairwise sums of the three numbers he thinks of; it appears that all six written numbers are distinct. Then King Arthur passes this leaf to Merlin. Can Merlin determine the three numbers King Arthur thinks of?

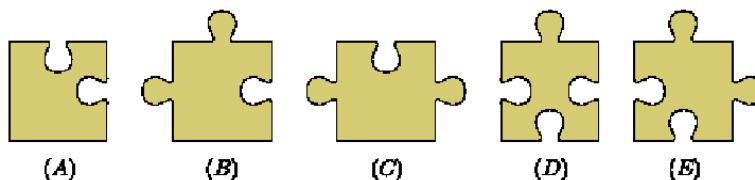
A8. Show that for arbitrary positive integers n, k there exists a polynomial $p(x)$, with degree at most $100\sqrt{nk}$, such that

$$p(0) > (|p(1)| + \dots + |p(n)|) + (|p(-1)| + \dots + |p(-k)|).$$

Combinatorics

C1. With his Excalibur, received from the Lady of the Lake (her arm clad in the purest shimmering silmte), King Arthur wants to free the town of Angnor from the fire-breathing dragon. The dragon has ~~five~~ three heads and ~~five~~ three tails. With one slash, he may cut off one head, two heads, one tail or two tails. If he cuts off one head, the dragon will immediately grow ~~five~~ three heads in its place. If he cuts off two heads, then nothing will grow in its place. If he cuts off one tail, there will be two tails growing immediately. Finally, if he cuts off two tails then the dragon will grow a head. What is the minimum number of cuts needed to kill the dragon (that is, in order that the dragon has no heads or tails remaining at all)?

C2. A jigsaw puzzle that says “850 pieces” actually consists of 851 pieces. Each piece is identical to one of the 5 samples shown in the *diagram*. How many pieces of type (E) are there in the puzzle?



C3. The Bank of Bath issues coins with an H on one side and a T on the other. Harry has n of these coins arranged in a line from left to right. He repeatedly performs the following operation: if there are exactly $k > 0$ coins showing H , then he turns over the k^{th} coin from the left; otherwise, all coins show T and he stops. For example, if $n = 3$ the process starting with the configuration THT would be $THT \rightarrow HHT \rightarrow HTT \rightarrow TTT$, which stops after three operations.

- Show that, for each initial configuration, Harry stops after a finite number of operations.
- For each initial configuration C , let $L(C)$ be the number of operations before Harry stops. For example, $L(THT) = 3$ and $L(TTT) = 0$. Determine the average value of $L(C)$ over all 2^n possible initial configurations C .

C4. A tournament of 49 competitors takes place, where every competitor plays one match against every other. A *cyclic three-variable inequality* consists of a set of three players $\{A, B, C\}$ such that A beats B , B beats C , and C beats A .

Géza would like there to be as many cyclic three-variable inequalities as possible at this tournament. How many can there be?

C5. A social network has 2019 users, some pairs of whom are friends. Whenever user A is friends with user B , user B is also friends with user A . Events of the following kind may happen repeatedly, one at a time:

Three users A , B , and C such that A is friends with both B and C , but B and C are not friends, change their friendship statuses such that B and C are now friends, but A is no longer friends with B , and no longer friends with C . All other friendship statuses are unchanged.

Initially, 1010 users have 1009 friends each, and 1009 users have 1010 friends each. Prove that there exists a sequence of such events after which each user is friends with at most one other user.

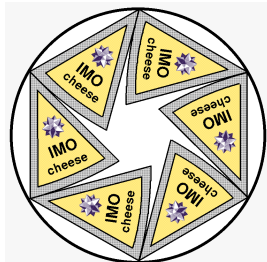
C6. An ancient King, who was not acquainted with an idea of the Round Table, summoned at some moment all his $n \geq 3$ knights, and tried to arrange them in a row in all possible ways. Surely, he had $n!$ ways, and he selected from them those with exactly k *disorders*, i.e. — k pairs of (not necessarily adjacent) knights such that the taller of them stands to the right of the shorter one. Assume that there were no two knights of the same height, that $k < \frac{n(n-1)}{4}$, and that Sir John is taller than Sir Robin. Prove that among the ways found by the King, there are at least a half where Sir John is to the left of Sir Robin.

C7. In the plane, marked are all points (x, y) where x and y are positive coprime integers. A knight moves along the grid lines by the following rule. He starts at some lattice point (x, y) , where $x > y \geq 1$, facing east. By a move, the knight reaches the nearest marked point at the direction it looks at, and then rotates 90° anticlockwise. Prove that the knight will eventually cycle if and only if he will never reach point $(1, 1)$.

C8. Let n be a positive integer, and let S be a family of $2n - 1$ open unit intervals (a_i, b_i) ($i = 1, 2, \dots, 2n - 1$) on the real line — so, $b_i = a_i + 1$ for all i . Take any $\varepsilon \in (0, 1/2)$ and mark all the points of the forms $a_i + \varepsilon$ and $b_i - \varepsilon$. Prove that there exists a marked point belonging to an odd number of intervals in S .

Geometry

G1. “IMO cheese” is a brand name of processed cheese that comes in a round box containing eight individually wrapped sectors that just fit in the box. Johnny ate two sectors for breakfast, and arranged the remaining ones in the box as shown in the figure. His Mom Pegasus did not like that. She said that Johnny was squeezing the cheese pieces. Johnny insisted that there was no squeezing, the pieces just fit. Who was right?



G2. Let F be the midpoint of circle arc AB , and let M be a point on the arc such that $AM > MB$. The perpendicular dropped from point F to AM intersects AM at point T . Show that T bisects the broken line AMB , that is $AT = TM + MB$.

G3. In triangle ABC , point A_1 lies on side BC and point B_1 lies on side AC . Let P and Q be points on segments AA_1 and BB_1 , respectively, such that PQ is parallel to AB . Let P_1 be a point on line PB_1 , such that B_1 lies strictly between P and P_1 , and $\angle PP_1C = \angle BAC$. Similarly, let Q_1 be a point on line QA_1 , such that A_1 lies strictly between Q and Q_1 , and $\angle CQ_1Q = \angle CBA$.

Prove that points P , Q , P_1 , and Q_1 are concyclic.

G4. Show that every closed curve c of length less than 2π on the surface of the unit sphere lies entirely on the surface of some hemisphere of the unit sphere.

G5. The circle k touches the circle ℓ internally at point P . A line p passing through P intersects the circles again at the points K and L , respectively. u is the tangent drawn at a point U of circle k . One intersection of u with the circle ℓ is V , and the intersection of lines KU and LV is T . Determine the locus of the point T as U traverses the circle k . (Consider both intersections of u and ℓ ; if $U = K$, the line KU is the tangent to k at K . Analogously, if $V = L$ then LV is the tangent drawn to ℓ at L .)

G6. A circle centred at I is tangent to the sides BC , CA , and AB of an acute-angled triangle ABC at A_1 , B_1 , and C_1 , respectively. Let K and L be the incenters of the quadrilaterals AB_1IC_1 and BA_1IC_1 , respectively. Let CH be an altitude of triangle ABC . Let the internal angle bisectors of angles AHC and BHC meet the lines A_1C_1 and B_1C_1 at P and Q , respectively. Prove that Q is the orthocenter of the triangle KLP .

G7. Let I be the incentre of acute triangle ABC with $AB \neq AC$. The incircle ω of ABC is tangent to sides BC , CA , and AB at D , E , and F , respectively. The line through D perpendicular to EF meets ω again at R . Line AR meets ω again at P . The circumcircles of triangles PCE and PBF meet again at Q .

Prove that lines DI and PQ meet on the line through A perpendicular to AI .

G8. Let two concentric circles k and K be given on the plane with radii r and R , $r < R$. Assume that the disc bounded by k is covered by angular domains whose vertices are within K . Then the sum of the angles of these angular domains is greater than or equal to the view angle of k from an arbitrary point of K .

G9. Do there exist incongruent convex quadrangular pyramids $SA_1A_2A_3A_4$ and $SB_1B_2B_3B_4$ such that, for every $i = 1, 2, 3, 4$, the triangles SA_iA_{i+1} and SB_iB_{i+1} are congruent (with correspondence of vertices)? Numeration of vertices is cyclic.

G10. Determine all positive integers $n \geq 3$ satisfying the following property:

Let X be a set of n points in general position in the plane. Then there exist several convex polygons $P_1, \dots, P_k$ such that no two of them have a common point (neither on the boundary, not in the interior), and the set of all vertices of the P_i coincides with X .

Number Theory

N1.

Find all pairs (k, n) of positive integers such that

$$k! = (2^n - 1)(2^n - 2)(2^n - 4) \cdots (2^n - 2^{n-1}).$$

N2.

Let a and n denote positive integers such that $a^n - 1$ is divisible by n . Prove that the numbers $a + 1, a^2 + 2, \dots, a^n + n$ all leave different remainders when divided by n .

N3.

Let a be a positive integer. Prove that there exists a positive integer b with the following property: For every prime p congruent to 1 modulo b , the residue a is not primitive modulo p . (In other words, for every such p with $p \nmid a$ there exists a positive integer $k < p - 1$ with $a^k \equiv 1 \pmod{p}$.)

N4.

Let $P(x)$, $Q(x)$ and $R(x, y)$ be three nonzero polynomials with integer coefficients. Distinct positive real numbers a and b satisfy $P(a) = Q(b) = 0$; however, for every positive integer n the numbers a^n and b^n are irrational. May it happen that $R(a^a, b^b) = 0$?

N5.

A sequence $a_1, a_2, \dots$ is determined by $a_1 = 7$ and $a_{n+1} = a_n + \gcd(n + 1, a_n)$ for all $n \geq 1$. Prove that for every n , the number $a_{n+1} - a_n$ has at most two positive divisors.

N6.

We say that a positive integer is *lazy* if it has no prime divisor greater than 3. Prove that there are at most two lazy numbers strictly between two consecutive square numbers.

N7.

Let n be a positive integer. In the Cartesian plane, consider the set X of all points with positive integer coordinates not exceeding n . Determine the largest cardinality of a subset $Y \subseteq X$ such that the distance between any two points in Y is integer.

N8.

Find positive constants a and c such that

$$\left| \sum_{k=1}^n \left\{ \frac{k^2}{N} \right\} - \frac{N}{2} \right| < cN^{1-a}$$

holds for all positive integers N .

Origin of the problems

- A1: IMO Shortlist 2019/A1; [IMO 2019/1](#); Liam Baker, South Africa
- A2: [KöMaL B. 4412](#). (January 2012); G. Kós, based on [Sam Lloyd's](#) puzzle
- A3: [KöMaL B. 5027](#). (April 2019); Attila Sztranyák
- A4: [KöMaL A. 744](#). (February 2019); based on Kürschák 2018/2
- A5: [KöMaL A. 732](#). (October 2018); related to [IMO 2018/2](#)
- A6: Kolmogorov Cup 2018, Major League, Day 3, Problem 3; I. Bogdanov
- A7: Kolmogorov Cup 2017, Major League, Day 4, Problem 2; A. Zaslavsky
- A8: [KöMaL A. 511](#). (May 2010); G. Kós
- C1: [KöMaL B. 4804](#). (September 2016; based on a German contest problem)
- C2: [KöMaL B. 4142](#). (January 2009); G. Kós
- C3: IMO Shortlist 2019/C3; [IMO 2019/5](#); David Altizio, USA
- C4: James Cranch
- C5: IMO Shortlist 2019/C5; [IMO 2019/3](#); Adrian Beker, Croatia
- C6: Kolmogorov Cup 2018, Major League, Day 1, Problem 2
- C7: Kolmogorov Cup 2017, Major League, Day 4, Problem 1
- C8: Kolmogorov Cup 2018, Major League, Day 4, Problem 6 [R. Pinchasi](#)
- G1: [KöMaL K. 124](#). (March 2007); G. Kós
- G2: KöMaL Gy. 2404. (March 1987); Archimedes of Syracuse
- G3: IMO Shortlist 2019/G3; [IMO 2019/2](#); Anton Trygub, Ukraine
- G4: [Miklós Schweitzer 1953/4](#)
- G5: [KöMaL B. 4511](#). (January 2013); András Hráskó
- G6: Kolmogorov Cup 2018, Major League, Day 3, Problem 1; A. Zaslavsky
- G7: IMO Shortlist 2019/G7; [IMO2019/6](#); Anant Mudgal, India
- G8: [Miklós Schweitzer 1985/3](#); Zsolt Páles
- G9: Kolmogorov Cup 2018, Major League, Day 4, Problem 9; L. Emelyanov, I. Bogdanov
- G10: Kolmogorov Cup 2018, Major League, Day 2, Problem 8; I. Bogdanov
- N1: IMO Shortlist 2019/N1; [IMO 2019/4](#); Gabriel Chicas Reyes, El Salvador
- N2: [KöMaL B. 4720](#). (May 2015)
- N3: Kolmogorov Cup 2018, Major League, Day 2, Problem 6
- N4: Kolmogorov Cup 2017, Major League, Day 4, Problem 5
- N5: Kolmogorov Cup 2013, Personal Olympiad, Seniors, Problem 7; E. S. Rowland
- N6: [KöMaL A. 717](#). (February 2018); Zoltán Gyenes
- N7: Kolmogorov Cup 2018, Major League, Day 2, Problem 1
- N8: [KöMaL A. 421](#). (February 2007)

Wanted back



At the end of IMO 2019, the knitted stellated dodecahedron had disappeared.
Please return it.

